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Units & Measurement

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MCQ (Single Correct Answer)	55 questions
Answer key	at the end

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MCQ (Single Correct Answer) 55 questions

Q1. RE-NEET 2026

Consider that σ_s, k_B, b represents Stefan-Boltzmann constant, Boltzmann constant and Wien's displacement law constant, respectively. The dimension of $\sigma_s k_B^{-1} b$ is

(A) $\left[\mathrm{L}^{-1}\,\mathrm{T}^{-1}\,\mathrm{K}^{-4}\right]$ (B) $\left[\mathrm{L}^{-1}\,\mathrm{T}^{-1}\,\mathrm{K}^{-2}\right]$

(C) $\left[\mathrm{L}^{-1}\,\mathrm{K}^{-2}\right]$ (D) $\left[\mathrm{L}^{-1}\,\mathrm{T}^{-1}\,\mathrm{K}^{-3}\right]$

Open solution

Q2. RE-NEET 2026

One main scale division of a Vernier calliper is equal to 1 mm and the number of divisions on the Vernier scale is 10 . When both the jaws touch each other, the Vernier scale shifts to the left of zero of the main scale in such a way that 4th Vernier division coincides with a division of the main scale. If this Vernier calliper measures the length of the wire to be 1 cm , the actual length of the wire is :

(A) 1.04 cm (B) 0.60 cm (C) 0.96 cm (D) 1.00 cm

Open solution

Q3. NEET 2026

The speed of light in vacuum is taken as unity. If light takes 6 min 40 s to reach the Earth from the Sun, the distance between the Sun and the Earth in new unit is:

(A) 3×10^8 (B) 500 (C) 3×10^{10} (D) 400

Open solution

Q4. NEET 2026

Each side of a metallic cube of mass 5.580 kg is measured to the 9.0 cm . Keeping the significant figures in view, the density of the material of the cube can be best expressed as $X \times 10^3\ \mathrm{kg\,m^{-3}}$ where the value of X is:

(A) 7.654 (B) 7.6 (C) 7.65 (D) 7.7

Open solution

Q5. NEET 2026

In a vernier calliper, 20 VSD coincide with 16 MSD (each division of length 1 mm). The least count of the vernier callipers is:

(A) 0.2 cm (B) 0.01 cm (C) 0.02 cm (D) 0.1 cm

Open solution

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Q6. NEET 2025

A balloon is made of a material of surface tension S and its inflation outlet (from where gas is filled in it) has small area A. It is filled with a gas of density ρ and takes a spherical shape of radius R. When the gas is allowed to flow freely out of it, its radius r changes from R to 0 (zero) in time T. If the speed $v(r)$ of gas coming out of the balloon depends on r as r^a and $T \propto S^\alpha A^\beta \rho^\gamma R^\delta$ then

Open solution

(A) $a = -\frac{1}{2}, \alpha = -\frac{1}{2}, \beta = -1, \gamma = \frac{1}{2}, \delta = \frac{7}{2}$

(B) $a = \frac{1}{2}, \alpha = \frac{1}{2}, \beta = -\frac{1}{2}, \gamma = \frac{1}{2}, \delta = \frac{7}{2}$

(C) $a = \frac{1}{2}, \alpha = \frac{1}{2}, \beta = -1, \gamma = +1, \delta = \frac{3}{2}$

(D) $a = -\frac{1}{2}, \alpha = -\frac{1}{2}, \beta = -1, \gamma = -\frac{1}{2}, \delta = \frac{5}{2}$

Q7. NEET 2025

A physical quantity P is related to four observations a, b, c and d as follows:

$P = a^3 b^2 / c\sqrt{d}$

The percentage errors of measurement in a, b, c and d are $1\%, 3\%, 2\%$, and 4% respectively. The percentage error in the quantity P is

Open solution

(A) 13% (B) 15% (C) 10% (D) 2%

Q8. NEET 2025

Consider the diameter of a spherical object being measured with the help of a Vernier callipers. Suppose its 10 Vernier Scale Divisions (V.S.D.) are equal to its 9 Main Scale Divisions (M.S.D.). The least division in the M.S. is 0.1 cm and the zero of V.S. is at $x = 0.1$ cm when the jaws of Vernier callipers are closed. If the main scale reading for the diameter is $M = 5$ cm and the number of coinciding vernier division is 8 , the measured diameter after zero error correction, is

Open solution

(A) 4.98 cm (B) 5.00 cm (C) 5.18 cm (D) 5.08 cm

Q9. NEET 2024 (Re-Examination)

In an electrical circuit, the voltage is measured as $V = (200 \pm 4)$ volt and the current is measured as $I = (20 \pm 0.2)$ A. The value of the resistance is:

Open solution

(A) $(10 \pm 4.2)\Omega$ (B) $(10 \pm 0.3)\Omega$ (C) $(10 \pm 0.1)\Omega$ (D) $(10 \pm 0.8)\Omega$

Q10. NEET 2024 (Re-Examination)

The pitch of an error free screw gauge is 1 mm and there are 100 divisions on the circular scale. While measuring the diameter of a thick wire, the pitch scale reads 1 mm and 63rd division on the circular scale coincides with the reference line. The diameter of the wire is:

Open solution

(A) 1.63 cm (B) 0.163 cm (C) 0.163 m (D) 1.63 m

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Q11. NEET 2024 (Re-Examination)

The potential energy of a particle moving along x-direction varies as $V = \frac{Ax^2}{\sqrt{x}+B}$. The dimensions of $\frac{A^2}{B}$ are:

(A) $\left[\mathrm{M}^{3/2}\,\mathrm{L}^{1/2}\,\mathrm{T}^{-3}\right]$

(B) $\left[\mathrm{M}^{1/2}\mathrm{LT}^{-3}\right]$

(C) $\left[M^2L^{1/2}T^{-4}\right]$

(D) $\left[\mathrm{ML}^2\,\mathrm{T}^{-4}\right]$

Open solution

Q12. NEET 2024

In a vernier callipers, $(N+1)$ divisions of vernier scale coincide with N divisions of main scale. If $1\ \mathrm{MSD}$ represents $0.1\ \mathrm{mm}$, the vernier constant (in cm) is:

(A) $\frac{1}{10\,\mathrm{N}}$

(B) $\frac{1}{100(N+1)}$

(C) $100\ \mathrm{N}$

(D) $10(N+1)$

Open solution

Q13. NEET 2024

The quantities which have the same dimensions as those of solid angle are:

(A) strain and angle

(B) stress and angle

(C) strain and arc

(D) angular speed and stress

Open solution

Q14. NEET 2024

A force defined by $F = \alpha t^2 + \beta t$ acts on a particle at a given time t. The factor which is dimensionless, if α and β are constants, is:

(A) $\beta t/\alpha$

(B) $\alpha t/\beta$

(C) $\alpha\beta t$

(D) $\alpha\beta/t$

Open solution

Q15. NEET 2023 Manipur

The diameter of a spherical bob, when measured with vernier callipers yielded the following values : $3.33\ \mathrm{cm}, 3.32\ \mathrm{cm}, 3.34\ \mathrm{cm}, 3.33\ \mathrm{cm}$ and $3.32\ \mathrm{cm}$. The mean diameter to appropriate significant figures is :

(A) $3.328\ \mathrm{cm}$

(B) $3.3\ \mathrm{cm}$

(C) $3.33\ \mathrm{cm}$

(D) $3.32\ \mathrm{cm}$

Open solution

Q16. NEET 2023 Manipur

The mechanical quantity, which has dimensions of reciprocal of mass (M^{-1}) is :

(A) angular momentum

(B) coefficient of thermal conductivity

(C) torque

(D) gravitational constant

Open solution

Q17. NEET 2023

The errors in the measurement which arise due to unpredictable fluctuations in temperature and voltage supply are :

(A) Personal errors

(B) Least count errors

(C) Random errors

(D) Instrumental errors

Open solution

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Q18. NEET 2023

A metal wire has mass (0.4 ± 0.002) g, radius (0.3 ± 0.001) mm and length (5 ± 0.02) cm. The maximum possible percentage error in the measurement of density will nearly be :

(A) 1.3% (B) 1.6% (C) 1.4% (D) 1.2%

Open solution

Q19. NEET 2022 Phase 2

The physical quantity that has the same dimensional formula as pressure is

(A) Coefficient of viscosity (B) Force
(C) Momentum (D) Young's modulus of elasticity

Open solution

Q20. NEET 2022 Phase 2

The percentage error in the measurement of g is : (Given that $g = \frac{4\pi^2 L}{T^2}$, $L = (10 \pm 0.1)$ cm, $T = (100 \pm 1)$ s)

(A) 7% (B) 2% (C) 5% (D) 3%

Open solution

Q21. NEET 2022 Phase 1

The dimensions $[MLT^{-2}A^{-2}]$ belong to the

(A) Magnetic flux (B) Self inductance
(C) Magnetic permeability (D) Electric permittivity

Open solution

Q22. NEET 2022 Phase 1

Plane angle and solid angle have

(A) Units but no dimensions (B) Dimensions but no units
(C) No units and no dimensions (D) Both units and dimensions

Open solution

Q23. NEET 2022 Phase 1

The area of a rectangular field (in m^2) of length 55.3 m and breadth 25 m after rounding off the value for correct significant digits is

(A) 138×10^1 (B) 1382 (C) 1382.5 (D) 14×10^2

Open solution

Q24. NEET 2021

If E and G respectively denote energy and gravitational constant, then $\frac{E}{G}$ has the dimensions of :

(A) $[M^2][L^{-2}][T^{-1}]$ (B) $[M^2][L^{-1}][T^0]$ (C) $[M][L^{-1}][T^{-1}]$ (D) $[M][L^0][T^0]$

Open solution

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Q25. NEET 2021

A screw gauge gives the following readings when used to measure the diameter of a wire

Main scale reading : 0 mm

Circular scale reading : 52 divisions

Given that 1 mm on main scale corresponds to 100 divisions on the circular scale. The diameter of the wire from the above data is :

(A) 0.052 cm (B) 0.52 cm (C) 0.026 cm (D) 0.26 cm

Open solution

Q26. NEET 2021

If force [F], acceleration [A] and time [T] are chosen as the fundamental physical quantities. Find the dimensions of energy.

(A) [F] [A^{-1}] [T] (B) [F] [A] [T] (C) [F] [A] [T^2] (D) [F] [A] [T^{-1}]

Open solution

Q27. NEET 2020 Phase 1

A screw gauge has least count of 0.01 mm and there are 50 divisions in its circular scale. The pitch of the screw gauge is :

(A) 0.25 mm (B) 0.5 mm (C) 1.0 mm (D) 0.01 mm

Open solution

Q28. NEET 2020 Phase 1

Taking into account of the significant figures, what is the value of $9.99m - 0.0099m$?

(A) 9.98 m (B) 9.980 m (C) 9.9 m (D) 9.9801 m

Open solution

Q29. NEET 2020 Phase 1

Dimensions of stress are :

(A) $[ML^2T^{-2}]$ (B) $[ML^0T^{-2}]$

(C) $[ML^{-1}T^{-2}]$ (D) $[MLT^{-2}]$

Open solution

Q30. NEET 2020 Phase 1

The energy required to break one bond in DNA is 10^{-20} J. This value in eV is nearly :

(A) 0.6 (B) 0.06 (C) 0.006 (D) 6

Open solution

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Q31. NEET 2019

In an experiment, the percentage of error occurred in the measurement of physical quantities A, B, C and D are 1%, 2 %, 3 % and 4 % respectively. Then the maximum percentage of error in the measurement X, where
X = $\frac{A^2B^{1/2}}{C^{1/3}D^3}$, will be :

Open solution

(A) 16%
(B) -10%
(C) 10%
(D) $\left(\frac{3}{13}\right)\%$

Q32. NEET 2018

A student measured the diameter of a small steel ball using a screw gauge of least count 0.001 cm. The main scale reading is 5 mm and zero of circular scale division coincides with 25 divisions above the reference level. If screw gauge has a zero error of −0.004 cm, the correct diameter of the ball is

Open solution

(A) 0.521 cm
(B) 0.525 cm
(C) 0.053 cm
(D) 0.529 cm

Q33. NEET 2017

A physical quantity of the dimensions of length that can be formed out of c, G and $\frac{e^2}{4\pi\varepsilon_0}$ is [c is velocity of light, G is the universal constant of gravitation and e is charge]

Open solution

(A) $c^2\left[G-\frac{e^2}{4\pi\varepsilon_0}\right]^{1/2}$
(B) $\frac{1}{c^2}\left[\frac{e^2}{G\,4\pi\varepsilon_0}\right]^{1/2}$
(C) $\frac{1}{c}G\frac{e^2}{4\pi\varepsilon_0}$
(D) $\frac{1}{c^2}\left[G\frac{e^2}{4\pi\varepsilon_0}\right]^{1/2}$

Q34. NEET 2016 Phase 2

Planck's constant (h), speed of light in vacuum (c) and Newton's gravitional constant (G) are three fundamental constants. Which of the following combinations of these has the dimension of length ?

Open solution

(A) $\frac{\sqrt{hG}}{c^{3/2}}$
(B) $\frac{\sqrt{hG}}{c^{5/2}}$
(C) $\sqrt{\frac{hc}{G}}$
(D) $\sqrt{\frac{Gc}{h^{3/2}}}$

Q35. AIPMT 2015

If dimensions of critical velocity v_c of a liquid flowing through a tube are expressed as $[\eta^x\rho^y r^z]$ where η, ρ and r are the coefficient of viscosity of liquid, density of liquid and radius of the tube respectively, then the values of x, y and z are given by

Open solution

(A) $-1,-1,-1$
(B) 1, 1, 1
(C) $1,-1,-1$
(D) $-1,-1,1$

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Q36. AIPMT 2015 Cancelled Paper

If energy (E), velocity (V) and time (T) are chosen as the fundamental quantities, the dimensional formula of surface tension will be

(A) $[EV^{-2}T^{-2}]$ (B) $[E^{-2}V^{-1}T^{-3}]$
(C) $[EV^{-2}T^{-1}]$ (D) $[EV^{-1}T^{-2}]$

Open solution

Q37. AIPMT 2014

If force (F), velocity (V) and time (T) are taken as fundamental units, then dimensions of mass are

(A) $[FVT^{-1}]$ (B) $[FVT^{-2}]$
(C) $[FV^{-1}T^{-1}]$ (D) $[FV^{-1}T]$

Open solution

Q38. NEET 2013 (Karnataka)

The pair of quantities having same dimensions is

(A) Impulse and Surface Tension (B) Angular momentum and Work
(C) Work and Torque (D) Young's modulus and Energy

Open solution

Q39. NEET 2013

If an experiment four quantities a, b, c and d are measured with percentage error 1%, 2%, 3% and 4% respectively.

Quantity P is calculated as follows P $= \frac{a^3b^2}{cd}$ % error in P is

(A) 7% (B) 4% (C) 14% (D) 10%

Open solution

Q40. AIPMT 2012 Mains

The dimensions of $(\mu_0\varepsilon_0)^{-1/2}$ are

(A) $[L^{1/2}T^{-1/2}]$ (B) $[L^{-1}T]$
(C) $[LT^{-1}]$ (D) $[L^{1/2}T^{1/2}]$

Open solution

Q41. AIPMT 2012 Prelims

The damping force on an oscillator is directly proportional to the velocity. The units of the constant of proportionality are

(A) kg m s^{-1} (B) kg m s^{-2} (C) kg s^{-1} (D) kg s

Open solution

Q42. AIPMT 2011 Mains

The density of a material in CGS system of units is 4 g cm^{-3}. In a system of units in which unit of length is 10 cm and unit of mass is 100 g, the value of density of material will be

(A) 0.04 (B) 0.4 (C) 40 (D) 400

Open solution

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Q43. AIPMT 2011 Prelims

The dimensions of $(\mu_0\varepsilon_0)^{-1/2}$ are

(A) $[L^{1/2}T^{-1/2}]$ (B) $[L^{-1}T]$

(C) $[LT^{-1}]$ (D) $[L^{1/2}T^{1/2}]$

Open solution

Q44. AIPMT 2010 Mains

A student measures the distance traversed in free fall of a body, initially at rest, in a given time. He uses this data to estimate g, the acceleration due to gravity. If the maximum percentage errors in measurement of the distance and the time are e_1 and e_2 respectively, the percentage error in the estimation of g is

(A) $e_2 - e_1$ (B) $e_1 + 2e_2$ (C) $e_1 + e_2$ (D) $e_1 - 2e_2$

Open solution

Q45. AIPMT 2010 Prelims

The dimension of $\frac{1}{2}\varepsilon_0 E^2$, where ε_0 is permittivity of free space and E is electric field, is

(A) ML^2T^{-2} (B) $ML^{-1}T^{-2}$ (C) ML^2T^{-1} (D) MLT^{-1}

Open solution

Q46. AIPMT 2009

If the dimensions of a physical quantity are given by $M^aL^bT^c$, then the physical quantity will be

(A) velocity if a = 1, b = 0, c = −1 (B) acceleration if a = 1, b = 1, c = −2

(C) force if a = 0, b = −1, c = −2 (D) pressure if a = 1, b = −1, c = −2

Open solution

Q47. AIPMT 2008

If the error in the measurement of radius of a sphere is 2%, then the error in the determination of volume of the sphere will be

(A) 8% (B) 2% (C) 4% (D) 6%

Open solution

Q48. AIPMT 2008

Which two of the following five physical parameters have the same dimensions ?

1. Energy density
2. Refractive index
3. Dielectric constant
4. Young's modulus
5. Magnetic field

(A) 1 and 4 (B) 1 and 5 (C) 2 and 4 (D) 3 and 5

Open solution

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Q49. AIPMT 2007

Dimensions of resistance in an electrical circuit, in terms of dimension of mass M, of length L, of time T and of current I, would be

(A) $[ML^2T^{-2}]$ (B) $[ML^2T^{-1}I^{-1}]$ (C) $[ML^2T^{-3}I^{-2}]$ (D) $[ML^2T^{-3}I^{-1}]$

Open solution

Q50. AIPMT 2006

The velocity v of a particle at time t is given by $v = at + \frac{b}{t+c}$, where a, b and c are constants. The dimensions of a, b and c are

(A) [L], [LT] and $[LT^{-2}]$
(B) $[LT^{-2}]$, [L] and [T]
(C) $[L^2]$, [T] and $[LT^{-2}]$
(D) $[LT^{-2}]$, and [LT] and [L]

Open solution

Q51. AIPMT 2005

The ratio of the dimensions of Planck's constant and that of moment of inertia is the dimensions of

(A) time
(B) frequency
(C) angular momentum
(D) velocity

Open solution

Q52. AIPMT 2004

The dimensions of universal gravitational constant are

(A) $[M^{-1}L^3T^{-2}]$ (B) $[ML^2T^{-1}]$ (C) $[M^{-2}L^3T^{-2}]$ (D) $[M^{-2}L^2T^{-1}]$

Open solution

Q53. AIPMT 2004

The unit of permittivity of free space, ε_0, is

(A) coulomb/newton-metre
(B) newton-metre^2/coulomb^2
(C) coulomb^2/newton-metre^2
(D) coulomb^2/(newton-metre)^2

Open solution

Q54. AIPMT 2001

The dimensions of Planck's constant equals to that of

(A) energy
(B) momentum
(C) angular momentum
(D) power

Open solution

Q55. AIPMT 2000

Which pair do not have equal dimensions ?

(A) Energy and torque
(B) Force and impulse
(C) Angular momentum and Planck constant
(D) Elastic modulus and pressure.

Open solution

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Answer Key

1. **B**	2. **A**	3. **D**	4. **D**	5. **C**
6. **A**	7. **A**	8. **A**	9. **B**	10. **B**
11. **C**	12. **B**	13. **A**	14. **B**	15. **C**
16. **D**	17. **C**	18. **B**	19. **D**	20. **D**
21. **C**	22. **A**	23. **D**	24. **B**	25. **A**
26. **C**	27. **B**	28. **A**	29. **C**	30. **B**
31. **A**	32. **D**	33. **D**	34. **A**	35. **C**
36. **A**	37. **D**	38. **C**	39. **C**	40. **C**
41. **C**	42. **C**	43. **C**	44. **B**	45. **B**
46. **D**	47. **D**	48. **A**	49. **C**	50. **B**
51. **B**	52. **A**	53. **C**	54. **C**	55. **B**

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